

Computer Vision Final Project

Vision Based Puzzle Solving

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Abstract:

The goal of the project is to utilize the classical computer vision technics to solve a puzzle given the solved puzzle image as a reference and the pieces to match. In this work what I did was to segment the pieces and extracted features from each of the individual pieces and the reference image, then tried to match the features to estimate an affine transform for each piece to properly place it on the canvas reconstructing the original image of the puzzle. The proposed method is first implemented on the simple PEPPA dataset resulting the perfect puzzle construction. Then it is generalized for the BLUE City dataset, and with some extra steps we managed to almost get the perfect reconstruction of 43 pieces out of 48.

Demonstration on PEPPA

Methodology:

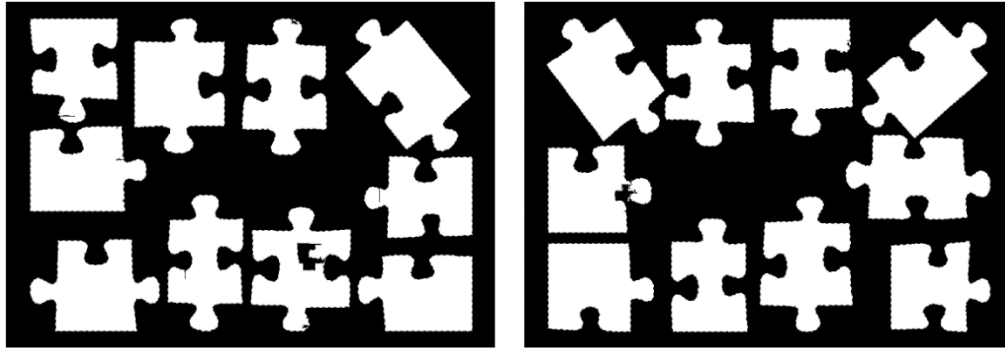
1. Loading Dataset

To start the process, first the images are imported the images for the main puzzle reference and the pieces using opencv imread. Since, the code is developed in a Jupyter Notebook, some functionalities are developed to show a single image or a grid of images (since we are dealing with large number of pieces in puzzles).

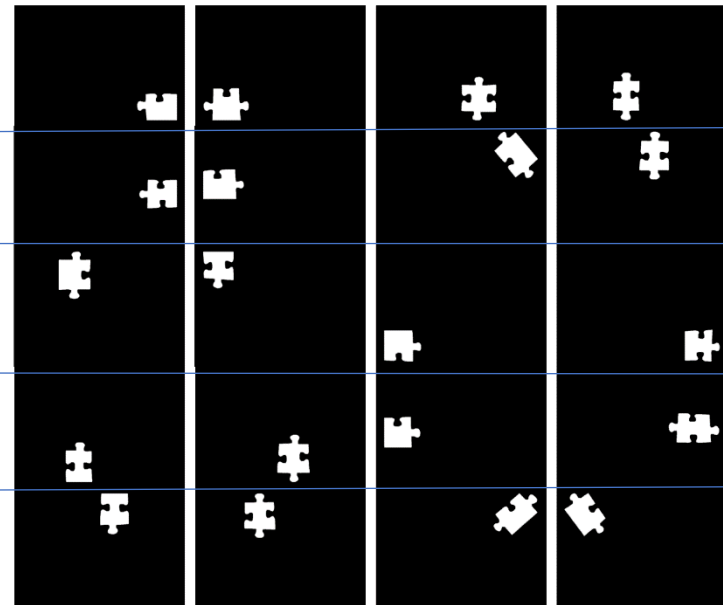


2. Piece Segmentation:

To start the segmentation process first a HSV range filter was applied to distinguish that background from the pieces. For the purpose, the image is converted into the HSV space and then a range filter is applied to produce a mask like follows:

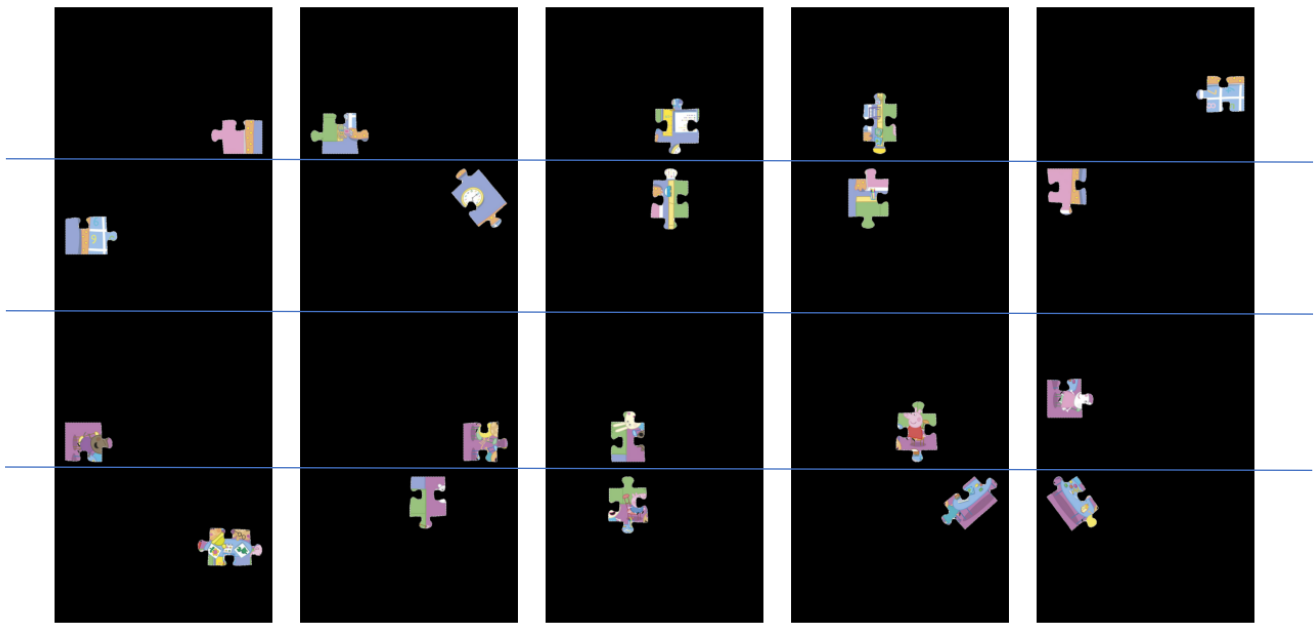


Obviously, to get the perfect background removal we need to get the perfect mask for the pieces. So, the mask is post processed to remove the imperfections (as small contours) and fill the cavities. Consecutively, as the perfect masks are created, a simple contour detection is applied to separate the masks for each piece and visualized as an image grid as follows:



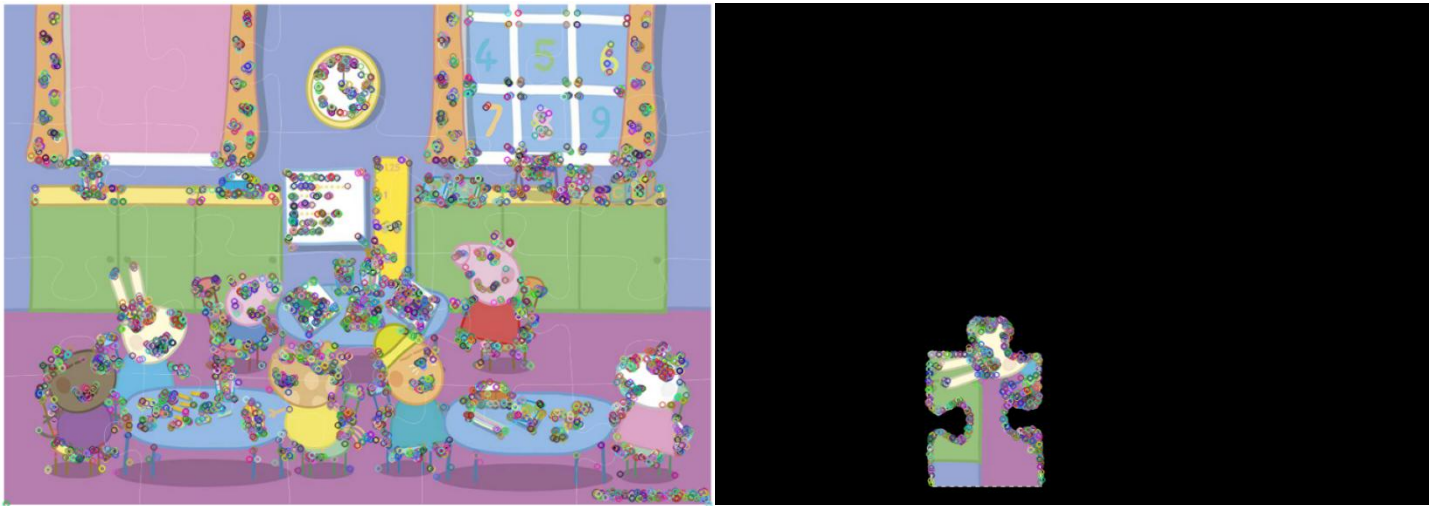
* Note that as the PEPPA puzzle has 20 pieces, this is a 5 by 4 grid of the pieces.

Having all the masks, the we can separate the pieces to start the feature processing part. The pieces segmented using a simple “bitwise and” operation to get the piece images ready for the feature extraction.



3. Feature Extraction and Matching:

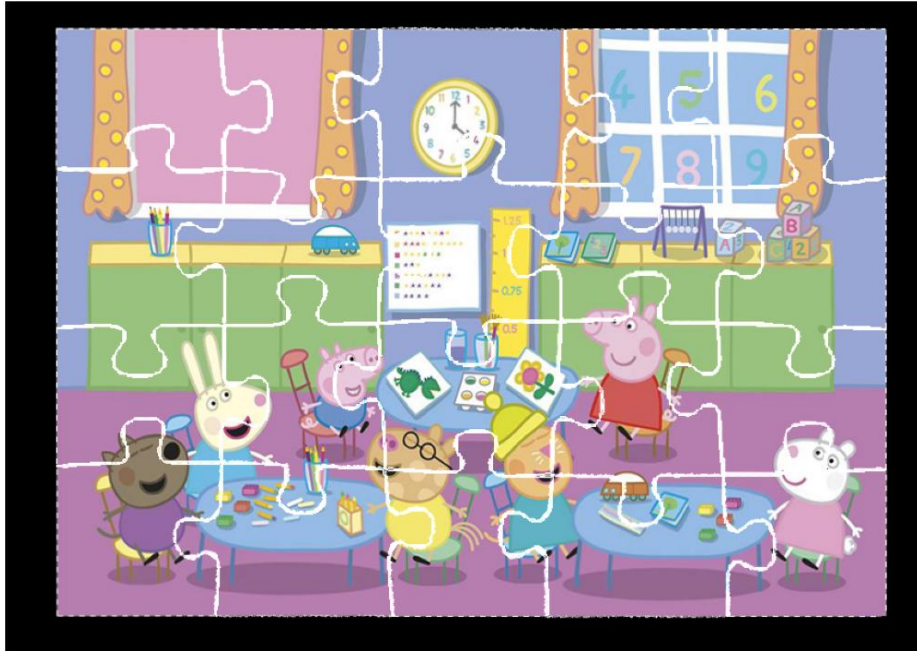
After having the segments ready, it is time to extract some features from the reference image and the puzzle pieces, to perform matching. For the purpose, ORB feature extractor is used. Resulting in:



Then a simple feature matching is applied to get the matches and by filtering the matched we can simply get matches that are capable of showing the rotation and displacement of the pieces. For the filtering a ratio test is applied. And matches created to create draw the affine transform matrix to rotate and place the image into its proper place.

4. Applying the Transforms to Reconstruct the Puzzle

Since affine transform matrices are created during the process of the feature matching everything is ready and only remaining is to create a black canvas and apply the transforms to the pieces, to rotate and place the pieces into the proper places. Resulting in:



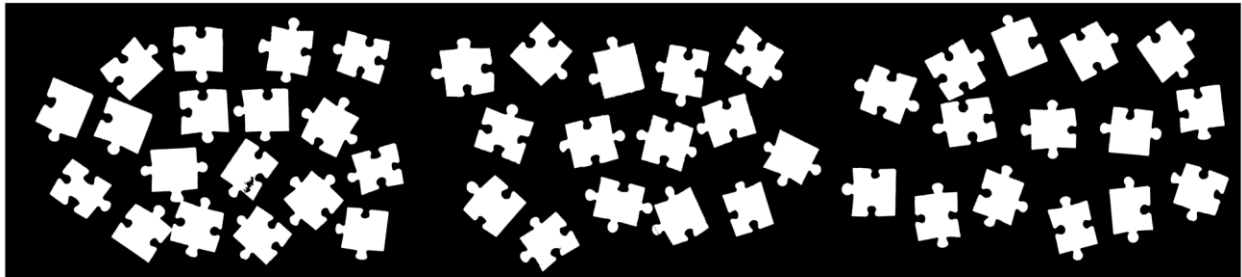
Since piece borders are imperfect, **alpha blending** method is also used to make the borders smoother resulting in:



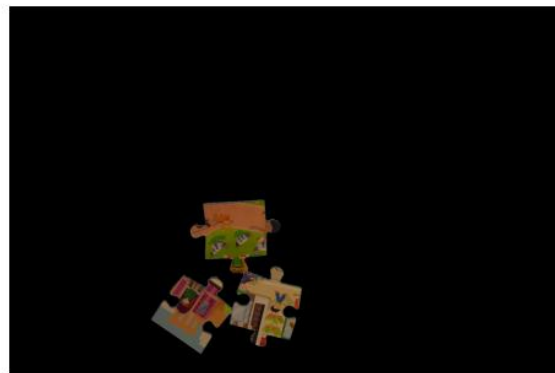
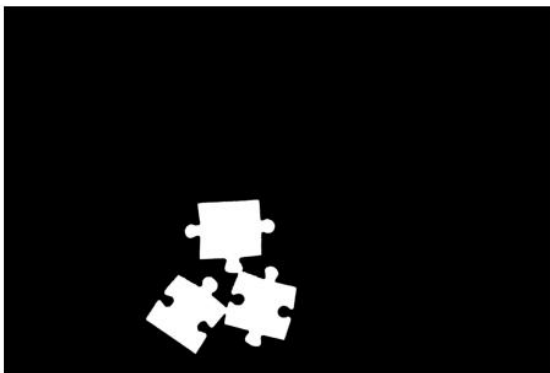
Demonstration on BLUE CITY

In this section, the implementation of the said methodology on the BLUE city is demoed.

First, an HSV range filter is applied to the puzzle pieces images, resulting in:

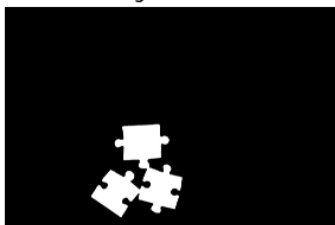


All the pieces are simply separated, except the cluster of three pieces in the first image that has connected contours as shown below:

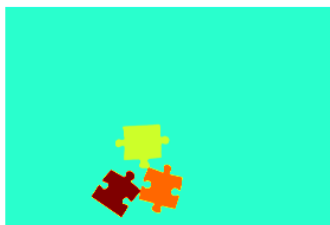


To separate the contours, the **water shedding algorithm** is implemented. By applying a distance transform on the masks and applying separation on the weakest points of the resulting image with pixels number smaller than a certain threshold. Which successfully separates the cluster into three:

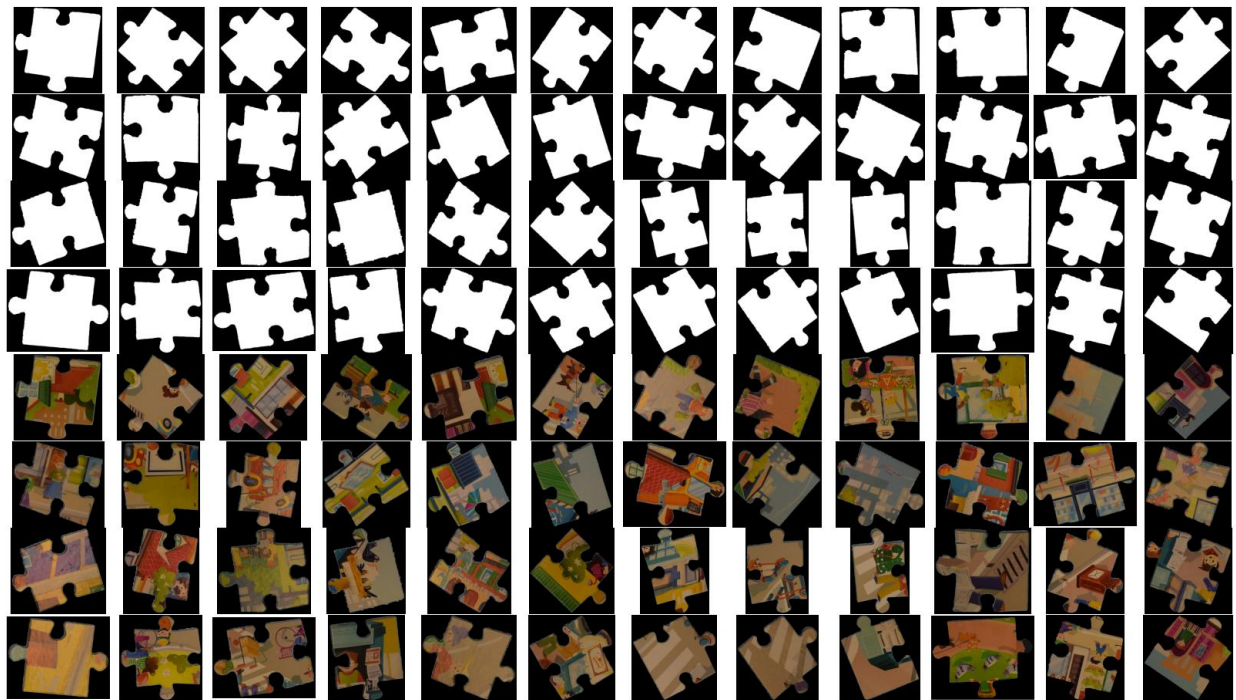
Original Mask



Markers



After this issue is also solved, the perfectly segmented pieces are available and we can proceed into piece separation and feature extraction.



Feature Extraction is again done using the ORB. However, to get better results before applying the feature extraction, the birds eye view of the reference image is extracted applying an HSV filter for background removal, corner detection and warp transform creation. Resulting in this:



Remaining parts are the same as the PEPPA, resulting as the reconstructed puzzle:



Unfortunately, five of the puzzle pieces didn't have good lighting conditions, resulting in poor features and matches. I couldn't handle the issue even after several attempts. However, 43 out of 48 pieces are placed properly in their places.

Conclusion

The proposed methods demonstrate a successful reconstruction of a puzzle implemented on two puzzle examples. For the feature extraction part SIFT feature extractor is also tested, but since ORB is a lot faster, this is what is kept in the procedure of the demo. Finally, the Alpha Blending method was tested which adds a lot in the smoothness of the final reconstructed image.